Landmark Papers in Machine Learning, Deep Learning, and Computer Vision

A Curated Chronological & Thematic Roadmap (100 Papers with DOIs)

Roadmap for Advanced Relearning & Research

23 de agosto de 2026

Este documento recopila 100 artículos pilares y revolucionarios de las áreas de *Machine Learning*, *Deep Learning* y *Computer Vision*.

1. Classical Machine Learning Foundations

1. **A Logical Calculus of the Ideas Immanent in Nervous Activity** (McCulloch & Pitts, 1943)
   https://doi.org/10.1007/BF02478259
2. **The Perceptron: A Probabilistic Model for Information Storage and Organization in the Brain** (Rosenblatt, 1958)
   https://doi.org/10.1037/h0042519
3. **Receptive Fields of Single Neurons in the Cat's Striate Cortex** (Hubel & Wiesel, 1959)
   https://doi.org/10.1113/jphysiol.1959.sp006308
4. **Learning Representations by Back-Propagating Errors** (Rumelhart, Hinton, & Williams, 1986)
   https://doi.org/10.1038/323533a0
5. **Support-Vector Networks** (Cortes & Vapnik, 1995)
   https://doi.org/10.1007/BF00994018
6. **Random Forests** (Breiman, 2001)
   https://doi.org/10.1023/A:1010933404324
7. **A Decision-Theoretic Generalization of On-Line Learning and an Application to Boosting** (Freund & Schapire, 1997)
   https://doi.org/10.1006/jcss.1997.1504
8. **Greedy Function Approximation: A Gradient Boosting Machine** (Friedman, 2001)
   https://doi.org/10.1214/aos/1013203451
9. **A Tutorial on Support Vector Machines for Pattern Recognition** (Burges, 1998)
   https://doi.org/10.1023/A:1009715923555
10. **Theory and Use of the EM Algorithm** (Gupta & Chen, 2010)
    https://doi.org/10.1561/2000000022

2. Classic Computer Vision (Pre-Deep Learning)

11. **Neocognitron: A Self-organizing Neural Network Model for a Mechanism of Visual Pattern Recognition** (Fukushima, 1980)
    https://doi.org/10.1007/BF00344251

12. **A Computational Approach to Edge Detection** (Canny, 1986)
    https://doi.org/10.1109/TPAMI.1986.4767851

13. **Determining Optical Flow** (Horn & Schunck, 1981)
    https://doi.org/10.1016/0004-3702(81)90024-2

14. **Distinctive Image Features from Scale-Invariant Keypoints (SIFT)** (Lowe, 2004)
    https://doi.org/10.1023/B:VISI.0000029664.99615.94

15. **Rapid Object Detection using a Boosted Cascade of Simple Features** (Viola & Jones, 2001)
    https://doi.org/10.1109/CVPR.2001.990517

16. **Histograms of Oriented Gradients for Human Detection (HOG)** (Dalal & Triggs, 2005)
    https://doi.org/10.1109/CVPR.2005.177

3. Deep Learning Foundations, Optimization & Regularization

17. **Gradient-Based Learning Applied to Document Recognition (LeNet-5)** (LeCun et al., 1998)
    https://doi.org/10.1109/5.726791

18. **A Fast Learning Algorithm for Deep Belief Nets** (Hinton, Osindero, & Teh, 2006)
    https://doi.org/10.1162/neco.2006.18.7.1527

19. **Understanding the Difficulty of Training Deep Feedforward Neural Networks** (Glorot & Bengio, 2010)
    https://doi.org/10.48550/arXiv.1002.4097

20. **Deep Sparse Rectifier Neural Networks (ReLU)** (Glorot, Bordes, & Bengio, 2011)
    https://proceedings.mlr.press/v15/glorot11a.html

21. **Dropout: A Simple Way to Prevent Neural Networks from Overfitting** (Srivastava et al., 2014)
    https://jmlr.org/papers/v15/srivastava14a.html

22. **Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift** (Ioffe & Szegedy, 2015)
    https://doi.org/10.48550/arXiv.1502.03167

23. **Adam: A Method for Stochastic Optimization** (Kingma & Ba, 2014)
    https://doi.org/10.48550/arXiv.1412.6980

24. **Delving Deep into Rectifiers: Surpassing Human-Level Performance on ImageNet Classification** (He et al., 2015)
    https://doi.org/10.1109/ICCV.2015.123

25. **Layer Normalization** (Ba, Kiros, & Hinton, 2016)
    https://doi.org/10.48550/arXiv.1607.06450

26. **Group Normalization** (Wu & He, 2018)
    https://doi.org/10.1007/978-3-030-01261-8_1

27. **The Lottery Ticket Hypothesis: Finding Sparse, Trainable Neural Networks** (Frankle & Carbin, 2018)
    https://doi.org/10.48550/arXiv.1803.03635

4. Convolutional Neural Networks (CNN Architectures)

28. **ImageNet Classification with Deep Convolutional Neural Networks (AlexNet)** (Krizhevsky, Sutskever, & Hinton, 2012)
https://doi.org/10.1145/3065386

29. **Visualizing and Understanding Convolutional Networks (ZFNet)** (Zeiler & Fergus, 2014)
https://doi.org/10.1007/978-3-319-10590-1_53

30. **Very Deep Convolutional Networks for Large-Scale Image Recognition (VGG)** (Simonyan & Zisserman, 2014)
https://doi.org/10.48550/arXiv.1409.1556

31. **Going Deeper with Convolutions (GoogLeNet/Inception)** (Szegedy et al., 2015)
https://doi.org/10.1109/CVPR.2015.7298594

32. **Deep Residual Learning for Image Recognition (ResNet)** (He et al., 2016)
https://doi.org/10.1109/CVPR.2016.90

33. **Densely Connected Convolutional Networks (DenseNet)** (Huang et al., 2017)
https://doi.org/10.1109/CVPR.2017.243

34. **MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications** (Howard et al., 2017)
https://doi.org/10.48550/arXiv.1704.04861

35. **Aggregated Residual Transformations for Deep Neural Networks (ResNeXt)** (Xie et al., 2017)
https://doi.org/10.1109/CVPR.2017.162

36. **EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks** (Tan & Le, 2019)
https://doi.org/10.48550/arXiv.1905.11946

37. **A ConvNet for the 2020s (ConvNeXt)** (Liu et al., 2022)
https://doi.org/10.1109/CVPR52688.2022.01167

5. Object Detection, Instance & Semantic Segmentation

38. **Rich Feature Hierarchies for Accurate Object Detection and Semantic Segmentation (R-CNN)** (Girshick et al., 2014)
https://doi.org/10.1109/CVPR.2014.81

39. **Fully Convolutional Networks for Semantic Segmentation (FCN)** (Long, Shelhamer, & Darrell, 2015)
https://doi.org/10.1109/CVPR.2015.7298965

40. **Fast R-CNN** (Girshick, 2015)
https://doi.org/10.1109/ICCV.2015.169

41. **Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks** (Ren et al., 2015)
https://doi.org/10.1109/TPAMI.2016.2577031

42. **U-Net: Convolutional Networks for Biomedical Image Segmentation** (Ronneberger, Fischer, & Brox, 2015)
https://doi.org/10.1007/978-3-319-24574-4_28

43. **You Only Look Once: Unified, Real-Time Object Detection (YOLO)** (Redmon et al., 2016)
https://doi.org/10.1109/CVPR.2016.91

44. **SSD: Single Shot MultiBox Detector** (Liu et al., 2016)
https://doi.org/10.1007/978-3-319-46448-0_2

45. **Feature Pyramid Networks for Object Detection (FPN)** (Lin et al., 2017)
https://doi.org/10.1109/CVPR.2017.106

46. **Focal Loss for Dense Object Detection (RetinaNet)** (Lin et al., 2017)
https://doi.org/10.1109/ICCV.2017.322

47. **Mask R-CNN** (He et al., 2017)
https://doi.org/10.1109/ICCV.2017.322

48. **End-to-End Object Detection with Transformers (DETR)** (Carion et al., 2020)
https://doi.org/10.1007/978-3-030-58452-8_13

6. Sequence Modeling, Attention & Transformer Architectures

49. **Long Short-Term Memory (LSTM)** (Hochreiter & Schmidhuber, 1997)
https://doi.org/10.1162/neco.1997.9.8.1735

50. **Empirical Evaluation of Gated Recurrent Neural Networks on Sequence Modeling (GRU)** (Chung et al., 2014)
https://doi.org/10.48550/arXiv.1412.3555

51. **Neural Machine Translation by Jointly Learning to Align and Translate** (Bahdanau, Cho, & Bengio, 2014)
https://doi.org/10.48550/arXiv.1409.0473

52. **Attention Is All You Need** (Vaswani et al., 2017)
https://doi.org/10.48550/arXiv.1706.03762

53. **BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding** (Devlin et al., 2018)
https://doi.org/10.48550/arXiv.1810.04805

54. **Language Models are Unsupervised Multitask Learners (GPT-2)** (Radford et al., 2019)
https://doi.org/10.48550/arXiv.1911.02116

55. **Language Models are Few-Shot Learners (GPT-3)** (Brown et al., 2020)
https://doi.org/10.48550/arXiv.2005.14165

7. Vision Transformers (ViTs) & Modern Visual Encoders

56. **An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale (ViT)** (Dosovitskiy et al., 2020)
https://doi.org/10.48550/arXiv.2010.11929

57. **Training Data-Efficient Image Transformers & Distillation Through Attention (DeiT)** (Touvron et al., 2021)
https://doi.org/10.48550/arXiv.2012.12877

58. **Swin Transformer: Hierarchical Vision Transformer using Shifted Windows** (Liu et al., 2021)
https://doi.org/10.1109/ICCV48922.2021.00986

59. **Masked Autoencoders Are Scalable Vision Learners (MAE)** (He et al., 2022)
https://doi.org/10.1109/CVPR52688.2022.01553

60. **Swin Transformer V2: Scaling Up Capacity and Resolution** (Liu et al., 2022)
https://doi.org/10.1109/CVPR52688.2022.01170

8. Self-Supervised Learning & Representation Learning

61. **Unsupervised Visual Representation Learning by Context Prediction** (Doersch, Gupta, & Efros, 2015)
https://doi.org/10.1109/ICCV.2015.168

62. **Representation Learning with Contrastive Predictive Coding (CPC)** (Oord, Li, & Vinyals, 2018)
https://doi.org/10.48550/arXiv.1807.03748

63. **A Simple Framework for Contrastive Learning of Visual Representations (SimCLR)** (Chen et al., 2020)
https://doi.org/10.48550/arXiv.2002.05709

64. **Momentum Contrast for Unsupervised Visual Representation Learning (MoCo)** (He et al., 2020)
https://doi.org/10.1109/CVPR42600.2020.00975

65. **Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning (BYOL)** (Grill et al., 2020)
https://doi.org/10.48550/arXiv.2006.07733

66. **Emerging Properties in Self-Supervised Vision Transformers (DINO)** (Caron et al., 2021)
https://doi.org/10.1109/ICCV48922.2021.00951

67. **DINOv2: Learning Robust Visual Features without Supervision** (Oquab et al., 2023)
https://doi.org/10.48550/arXiv.2304.07193

9. Generative Models (VAEs, GANs, Flow & Diffusion)

68. **Auto-Encoding Variational Bayes (VAE)** (Kingma & Welling, 2013)
https://doi.org/10.48550/arXiv.1312.6114

69. **Generative Adversarial Nets (GAN)** (Goodfellow et al., 2014)
https://doi.org/10.1145/3422622

70. **Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks (DCGAN)** (Radford, Metz, & Chintala, 2015)
https://doi.org/10.48550/arXiv.1511.06434

71. **Conditional Generative Adversarial Nets (cGAN)** (Mirza & Osindero, 2014)
https://doi.org/10.48550/arXiv.1411.1784

72. **Image-to-Image Translation with Conditional Adversarial Networks (Pix2Pix)** (Isola et al., 2017)
https://doi.org/10.1109/CVPR.2017.632

73. **Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks (CycleGAN)** (Zhu et al., 2017)
https://doi.org/10.1109/ICCV.2017.244

74. **A Style-Based Generator Architecture for Generative Adversarial Networks (StyleGAN)** (Karras, Laine, & Aila, 2019)
https://doi.org/10.1109/CVPR.2019.00453

75. **Deep Unsupervised Learning using Nonequilibrium Thermodynamics (Diffusion Models)** (Sohl-Dickstein et al., 2015)
https://doi.org/10.48550/arXiv.1503.03585

76. **Denoising Diffusion Probabilistic Models (DDPM)** (Ho, Jain, & Abbeel, 2020)
https://doi.org/10.48550/arXiv.2006.11239

77. **Score-Based Generative Modeling through Stochastic Differential Equations** (Song et al., 2020)
https://doi.org/10.48550/arXiv.2011.13456

78. **Classifier-Free Diffusion Guidance** (Ho & Salimans, 2021)
https://doi.org/10.48550/arXiv.2207.12598

79. **High-Resolution Image Synthesis with Latent Diffusion Models (Stable Diffusion)** (Rombach et al., 2022)
https://doi.org/10.1109/CVPR52688.2022.01042

80. **Flow Matching for Generative Modeling** (Lipman et al., 2022)
https://doi.org/10.48550/arXiv.2210.02747

10. Vision-Language & Multimodal Foundation Models

81. **Show and Tell: A Neural Image Caption Generator** (Vinyals et al., 2015)
https://doi.org/10.1109/CVPR.2015.7298935

82. **Show, Attend and Tell: Neural Image Caption Generation with Visual Attention** (Xu et al., 2015)
https://doi.org/10.48550/arXiv.1502.03044

83. **Learning Transferable Visual Models From Natural Language Supervision (CLIP)** (Radford et al., 2021)
https://doi.org/10.48550/arXiv.2103.00020

84. **BLIP: Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation** (Li et al., 2022)
https://doi.org/10.48550/arXiv.2201.12086

85. **BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models** (Li et al., 2023)
https://doi.org/10.48550/arXiv.2301.12597

86. **Visual Instruction Tuning (LLaVA)** (Liu et al., 2023)
https://doi.org/10.48550/arXiv.2304.08485

87. **Segment Anything (SAM)** (Kirillov et al., 2023)
https://doi.org/10.1109/ICCV51070.2023.00371

11. 3D Vision, Neural Rendering & Spatial AI

88. **PointNet: Deep Learning on Point Sets for 3D Classification and Segmentation** (Qi et al., 2017)
https://doi.org/10.1109/CVPR.2017.16

89. **PointNet++: Deep Hierarchical Feature Learning on Point Sets in a Metric Space** (Qi et al., 2017)
https://doi.org/10.48550/arXiv.1706.02413

90. **Occupancy Networks: Learning 3D Reconstruction in Function Space** (Mescheder et al., 2019)
https://doi.org/10.1109/CVPR.2019.00459

91. **NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis** (Mildenhall et al., 2020)
https://doi.org/10.1007/978-3-030-58452-8_24

92. **3D Gaussian Splatting for Real-Time Radiance Field Rendering** (Kerbl et al., 2023)
https://doi.org/10.1145/3592433

12. Foundational Physics, Geometry & Advanced Paradigms

93. **Group Equivariant Convolutional Networks** (Cohen & Welling, 2016)
https://doi.org/10.48550/arXiv.1602.07576

94. **Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges** (Bronstein et al., 2021)
https://doi.org/10.48550/arXiv.2104.13478

95. **Graph Attention Networks (GAT)** (Veličković et al., 2017)
https://doi.org/10.48550/arXiv.1710.10903

96. **Mastering the Game of Go with Deep Neural Networks and Tree Search (AlphaGo)** (Silver et al., 2016)
https://doi.org/10.1038/nature16961

97. **Highly Accurate Protein Structure Prediction with AlphaFold** (Jumper et al., 2021)
https://doi.org/10.1038/s41586-021-03819-2

98. **LoRA: Low-Rank Adaptation of Large Language Models** (Hu et al., 2021)
https://doi.org/10.48550/arXiv.2106.09685

99. **Training Compute-Optimal Large Language Models (Chinchilla)** (Hoffmann et al., 2022)
https://doi.org/10.48550/arXiv.2203.15556

100. **Mamba: Linear-Time Sequence Modeling with Selective State Spaces** (Gu & Dao, 2023)
https://doi.org/10.48550/arXiv.2312.00752